RECOMMENDATION SYSTEMS - 2

Agenda ⇒ Recap

→ DATASET , → COLLABORATIVE filtering

→ CONTENT BASED REC SYSTEM

items

Trans

support

N ⇒ $2^n - 1$ ⇒ min_support

10000

I items ⇒ $2^I - 1$

10 - 18

Ten.

Support ⇒

0 1 0 1
0 1 0 1

(Association Rules)

Apriori

which item sets are more frequent

(A, B)

(A, B, C)

ex

{milk, Bread} → {Jam, Eggs}

X → Y

Y → X

Bread → Jam

Jam → Bread ✗

Dominos

Pizza + Coke

Pizza → garlic Bread

{Pizza} → {Coke}

Antecedent

Consequent

Association rule mining

- Support
- Confidence
- Lift

support ⇒ min-support

item, item set

Txn

P(x in T) support(x) = X / N

confidence ⇒

Association Rules

X → Y

{milk, bread} → {jam, eggs}

Bread, jam

Butter

confidence

confidence (X → Y) = X ∩ Y / Txn X

P(Y|X)

confidence

Lift ⟹

Cornflakes — Milk

High conf.

Yogurt → Milk

TB → Milk ✗

Txn = 100

80 → milk

TB

10 → TB + Milk

TB → Milk

= 10 / 14 = 0.7

Lift ⟹ ✗

Lift (Toothbrush → Milk)

= Confidence

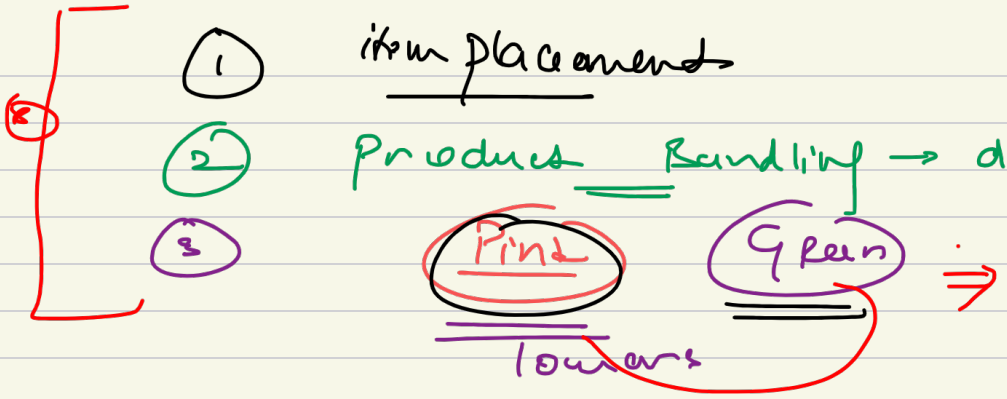
P(milk)

Lift < 1

Lift = 1

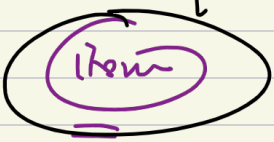
Lift > 1

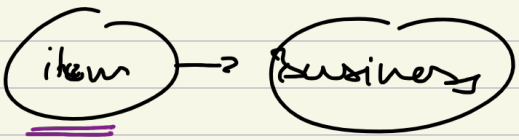
= 0.7 / 0.8 = 0.89

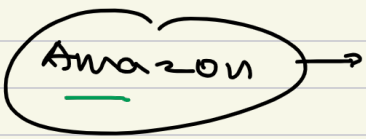

1
item placement
2
Products Bundling → discount
3
Pink
Green
Towers

④ Recommendation system





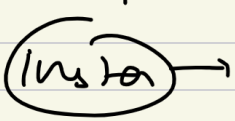




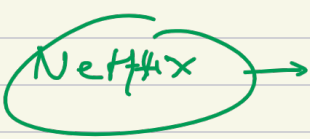
Products

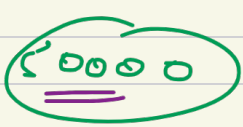
Netflix → movies

Youtube → Videos | Shorts

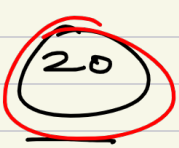


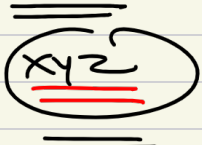
Reels | posts



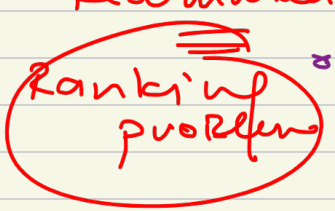


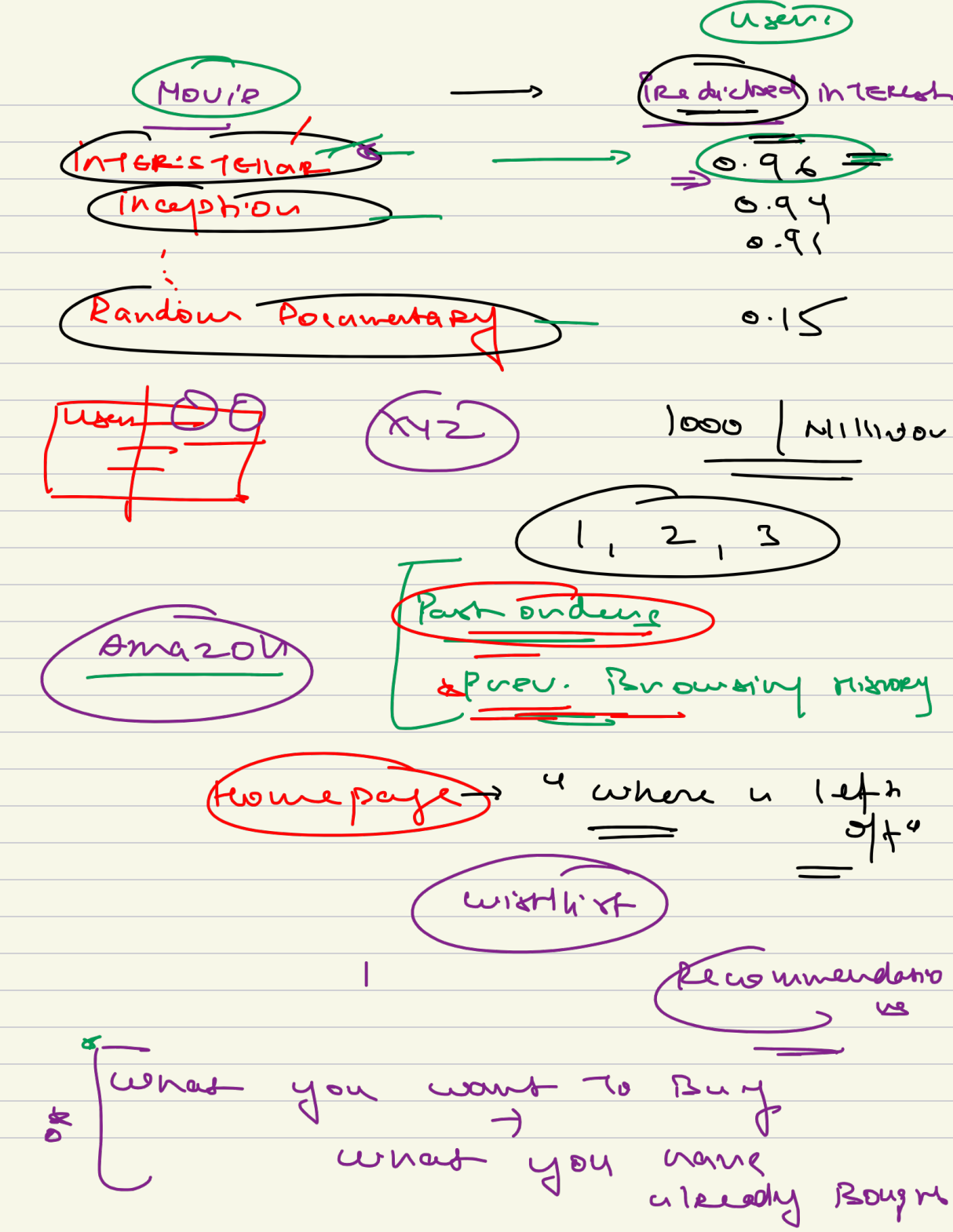
items



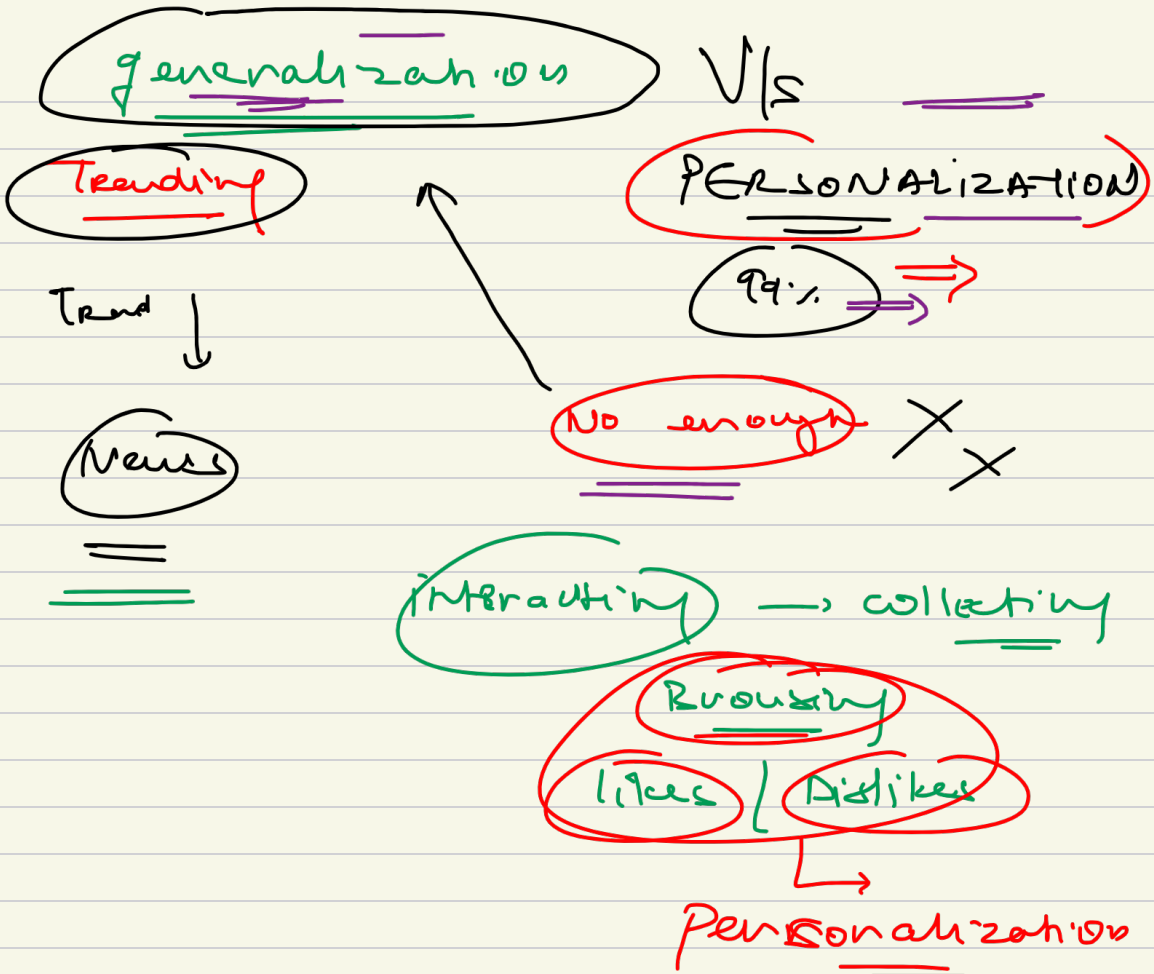


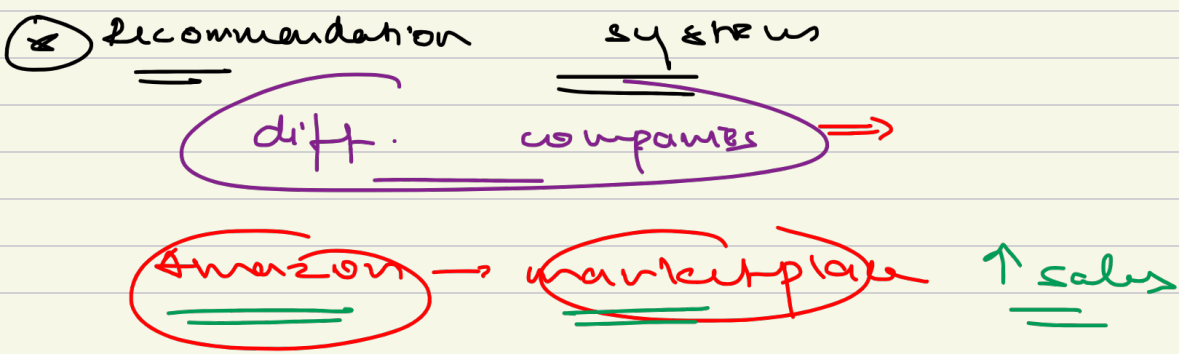
Recommendation

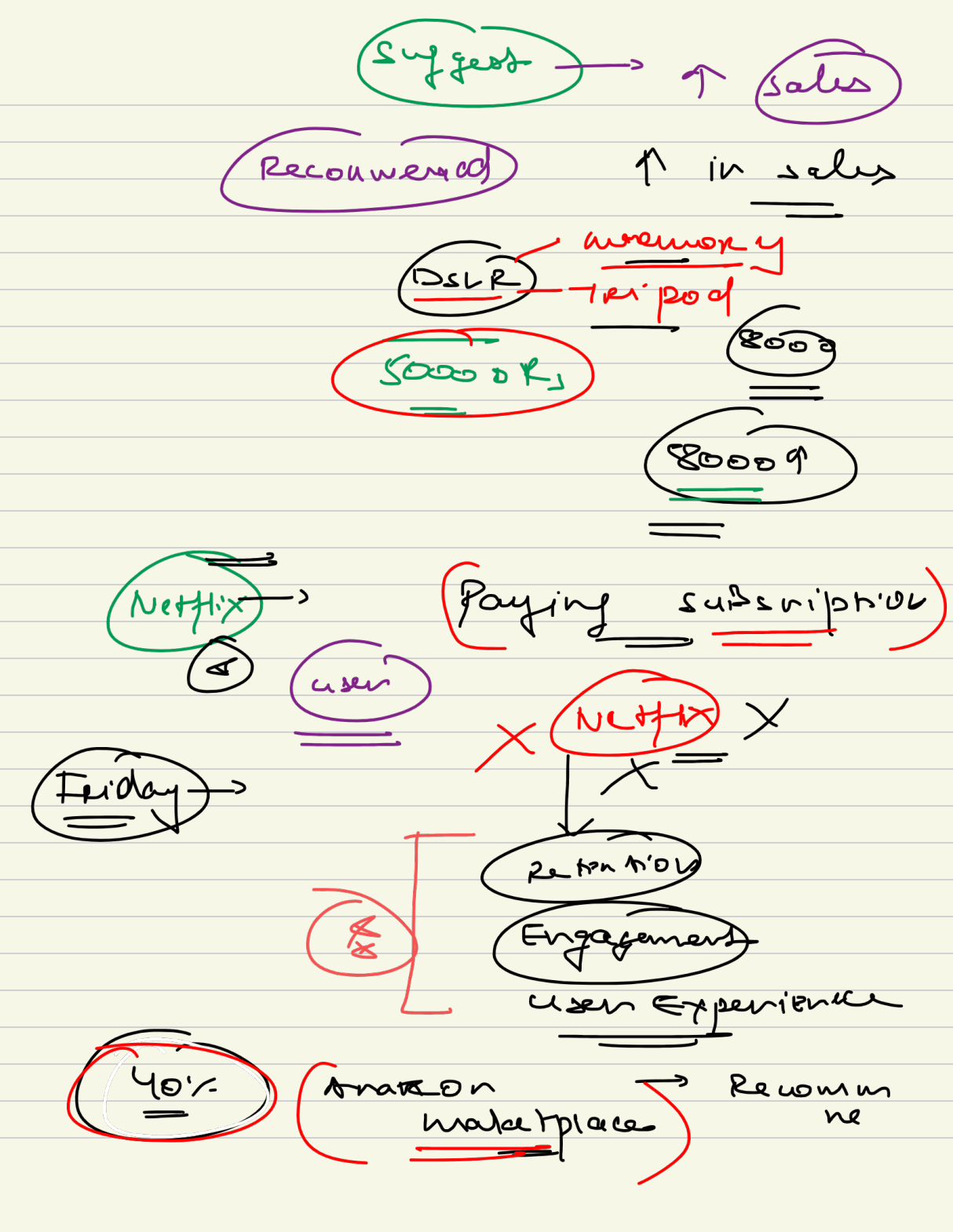



Users
Movie
Predicted interest
Interstellar
0.96
Inception
0.94
0.91
Random Documentary
0.15
User
XYZ
1000 | Millions
1, 2, 3
Amazon
Past ordering
& Prev. Browsing History
Homepage → " where u left off "
Wishlist
Recommendations
&
what you want to Buy
what you have already Bought


Amazon
Browsing history
mechanical keyboard
keyboards
MacBook
products related to this item
cleaners
USB
Youtube
New
accounts
suggestions
videos
Trending
loc
user1
user2


generalization
Vs
Trending
PERSONALIZATION
99%
Trend
News
No enough
interacting
collecting
Browsing
likes / dislikes
Personalization


Recommendation
systems
diff.
compambs
Amazon
marketplace
↑ sales


Suggest
Sales
Recommend
↑ in sales
DSLR
memory
tripod
$5000 DSLR
$800
$8000
Netflix
Paying subscription
user
Friday
NETFLIX
Retention
Engagement
User Experience
40%
Amazon
marketplace
Recommend

Rec system → ↑ + user Journey

System Design

ML

Recommendation system History

< 2007

Similarity collaborative content Based Ri~

simple

2008 – 2016

algo

SVD [(x) Matrix factorization]

10x ↑

> 2016

DATA ↑

⇒ Deep learning ✕

Collaborative ⇒

Embedding ⇒ NLP

Productions

Post a Recommendation

i/p o/p

Users : U_i $i: 1 \to n$

items : I_j $j: 1 \to m$

$n \gg m$

U_i → $I_{10}, I_{12}, I_{16}, I_4$

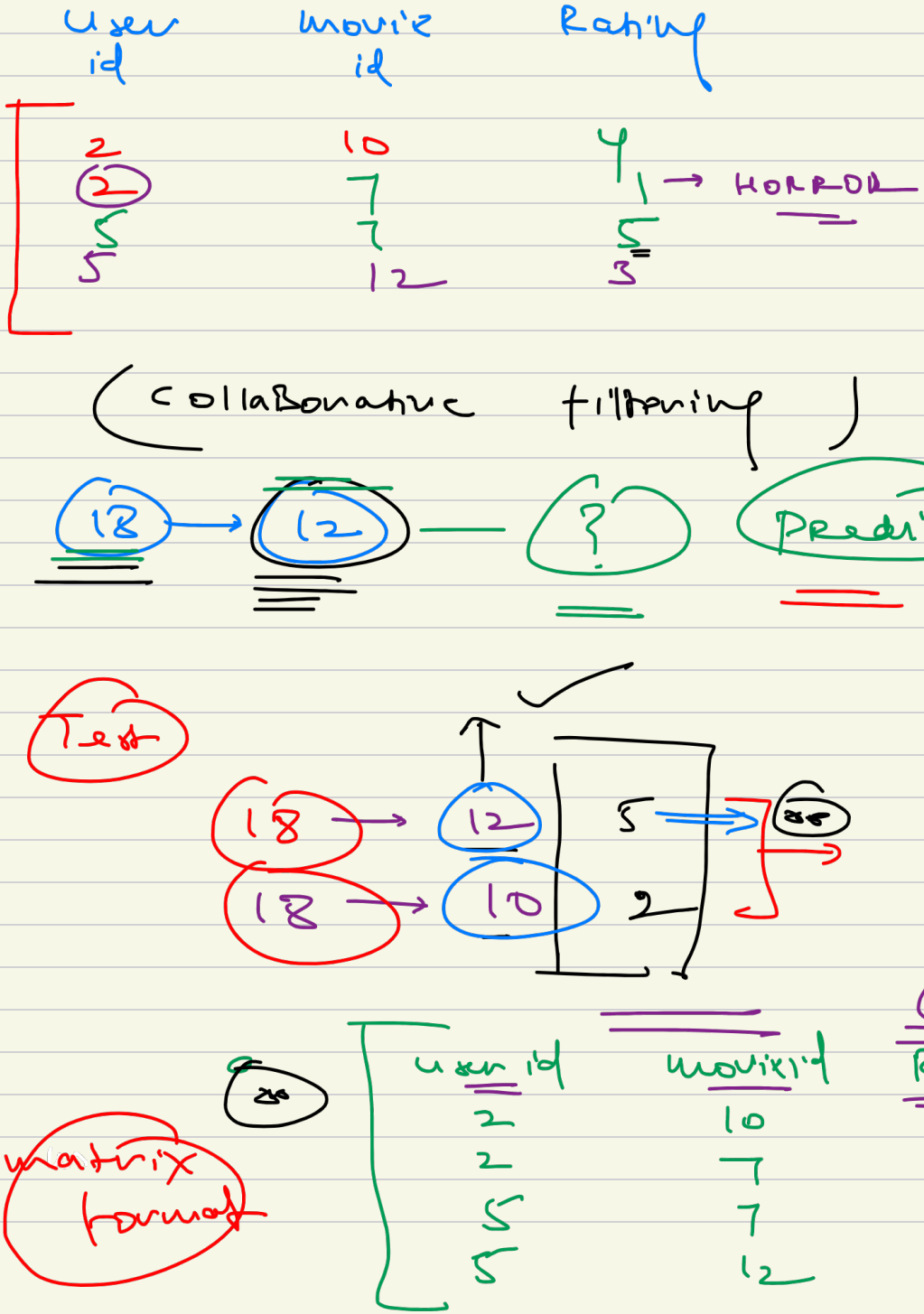
U_7

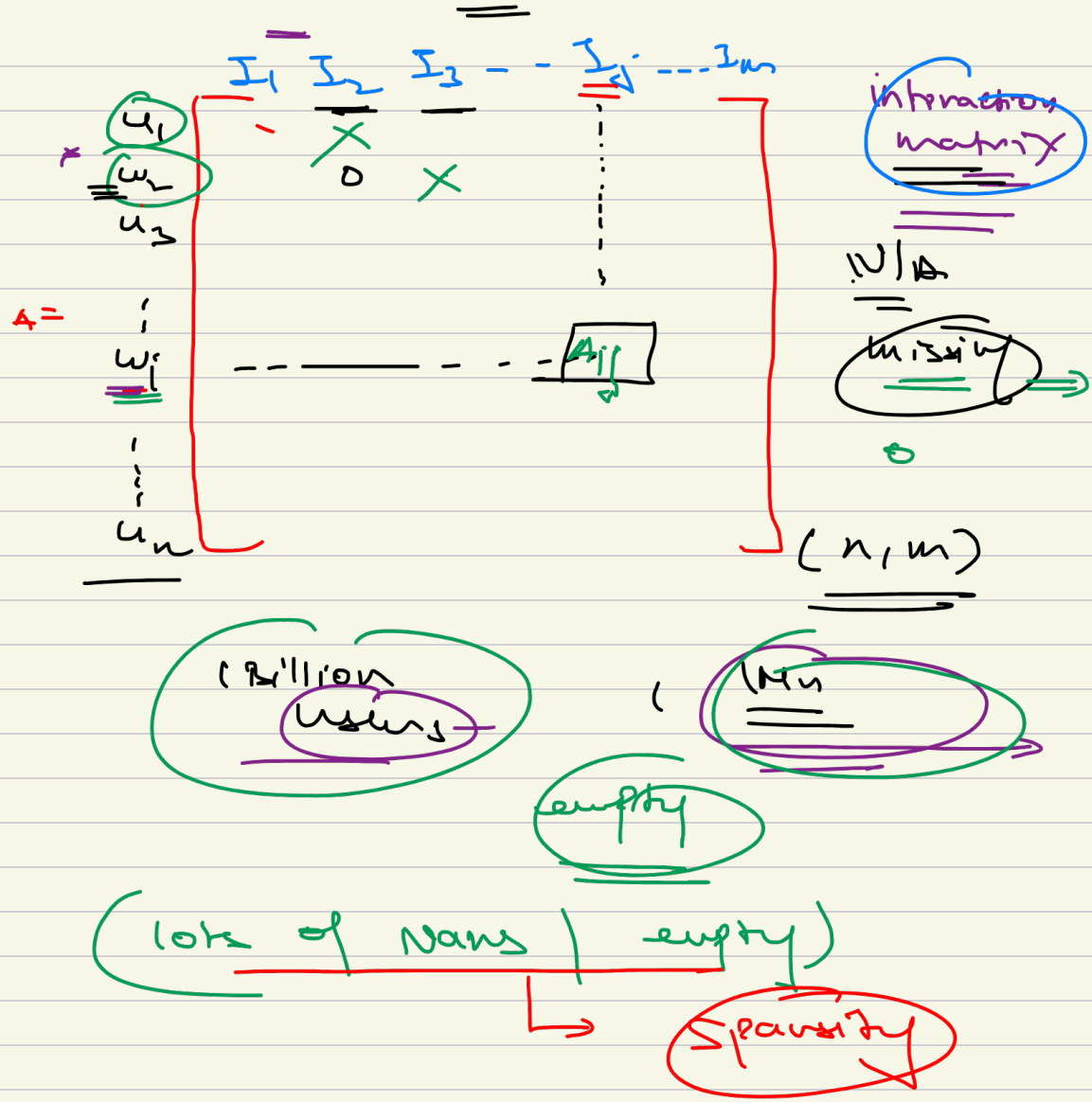
U_{18} → I_{12}

user watch History

users liked or not

user gave rating or not


User id
Movie id
Rating
2
2
5
5
10
7
7
12
4
1
5
3
HORROR
collaborative filtering
18
12
?
Predict
Test
18
18
12
10
5
2
matrix format
user id
movie id
Rating
2
2
5
5
10
7
7
12
4
1
5
2



Total cells = $n \times m$

$10^9 \times 10^6$

$= 10^{15}$ cells

non empty cells

1 ⟶ 100 movies

$10^9 \times 100 = 10^{11}$

Non empty cells

Non sparsity $= \frac{10^{11}}{10^{15}} = \frac{1}{10^4}$

$= 0.0001$

interaction matrix

99.99% sparsity

(user – user similarity)

Collaborative filtering

(item – item similarity)

Explicit | implicit

User → preferences

Infer the preferences from Behaviour

Netflix = 5 stars 4 star

90 min ✓

Song → 5 times

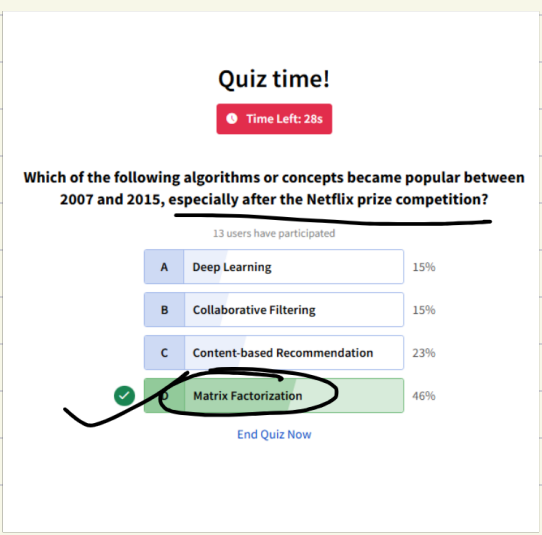
Amazon

Spotify ↑ ↓

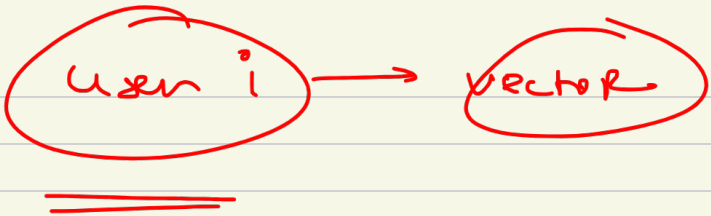
User → Bought ✓

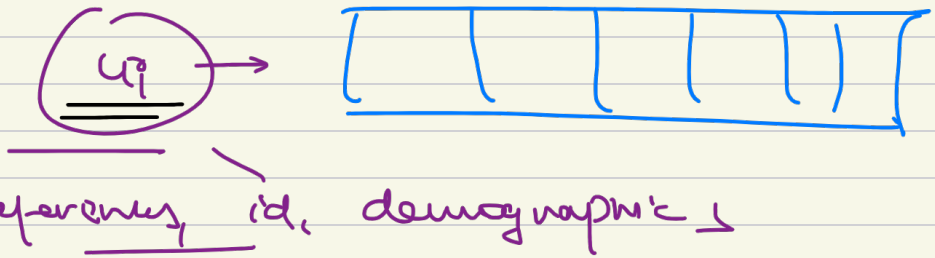
Reel → 0.5

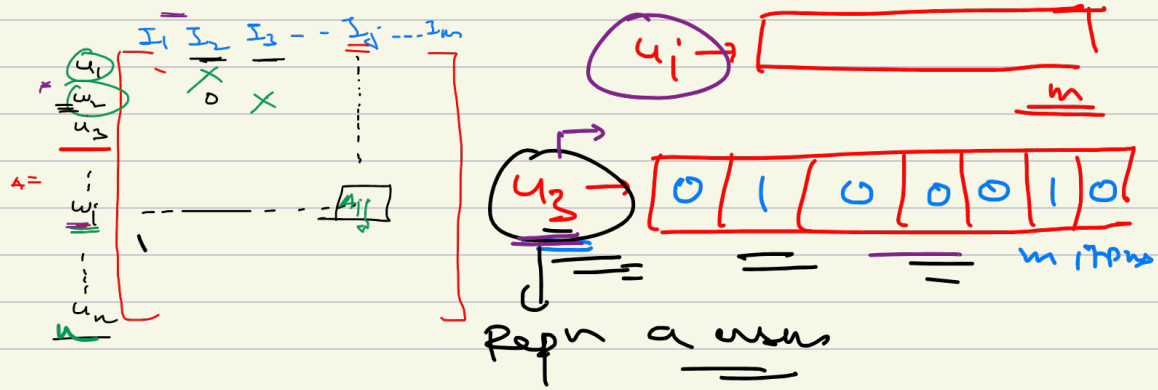
= Reel → 4.7 | 5

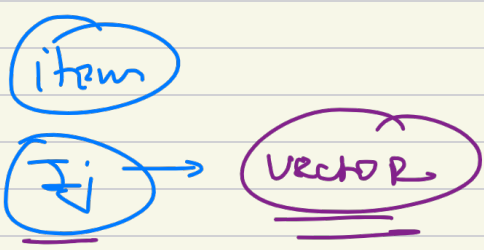


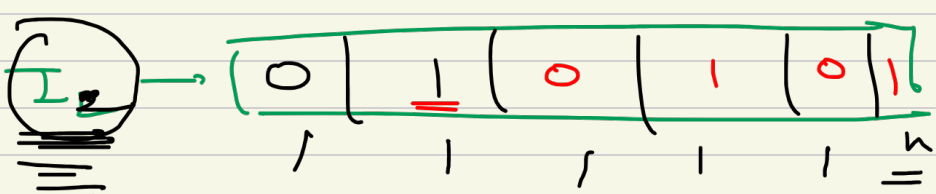
2006



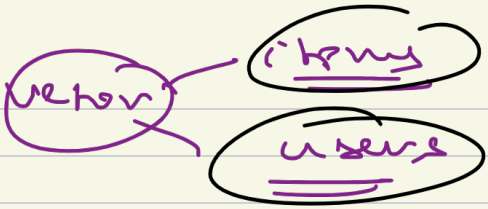








VECTORIZATION



item item similarity

U_0

Rating

I_4 = 0 0 5 2 0 0

U_0 I_5 =

I_8 =

t n

@sen

item item

I_4 , I_5 I_8

U_0

Similarity

cosine similarity

I_2

$$\text{Cosine similarity}(I_{12}, I_j) = \frac{I_{12} \cdot I_j}{\|I_{12}\| \, \|I_j\|}$$

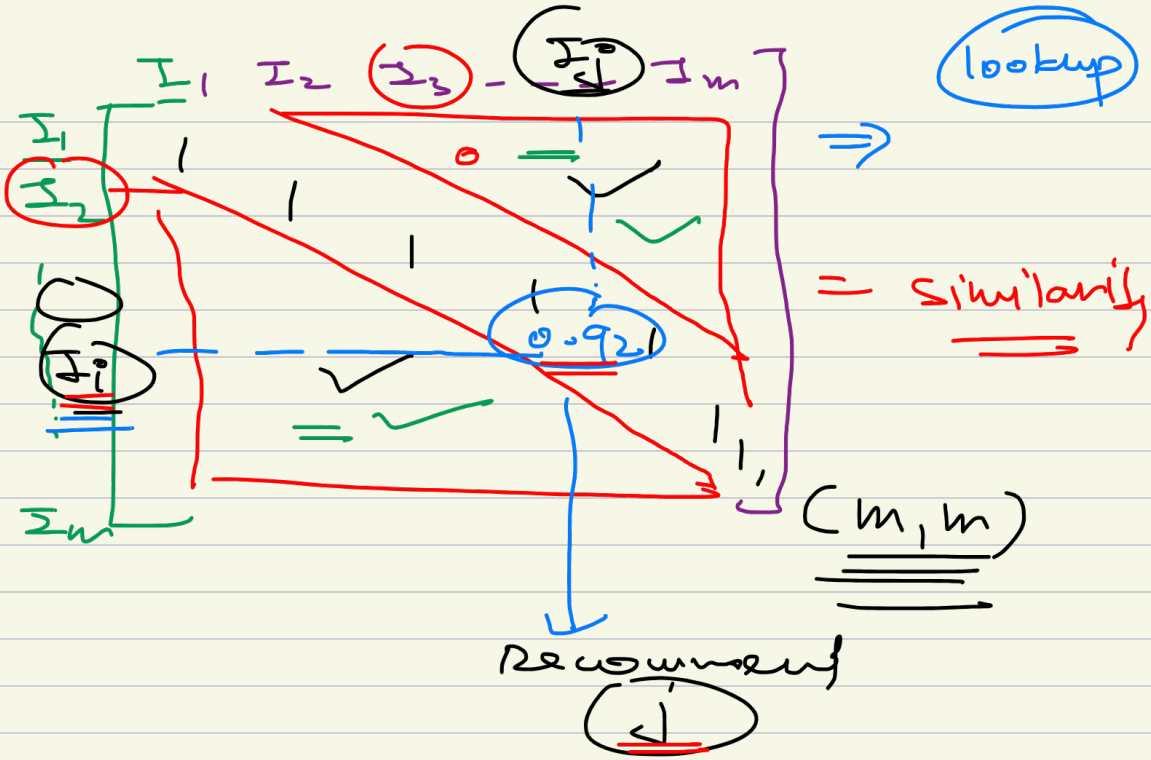
$u_{10} \Rightarrow$

$\cos(I_5, I_j)$

$\cos(I_8, I_j)$

cosine $\rightarrow [-1, 1]$

$\uparrow$ Recommend



I_3

I_2

$I_2 \rightarrow$

16 movies

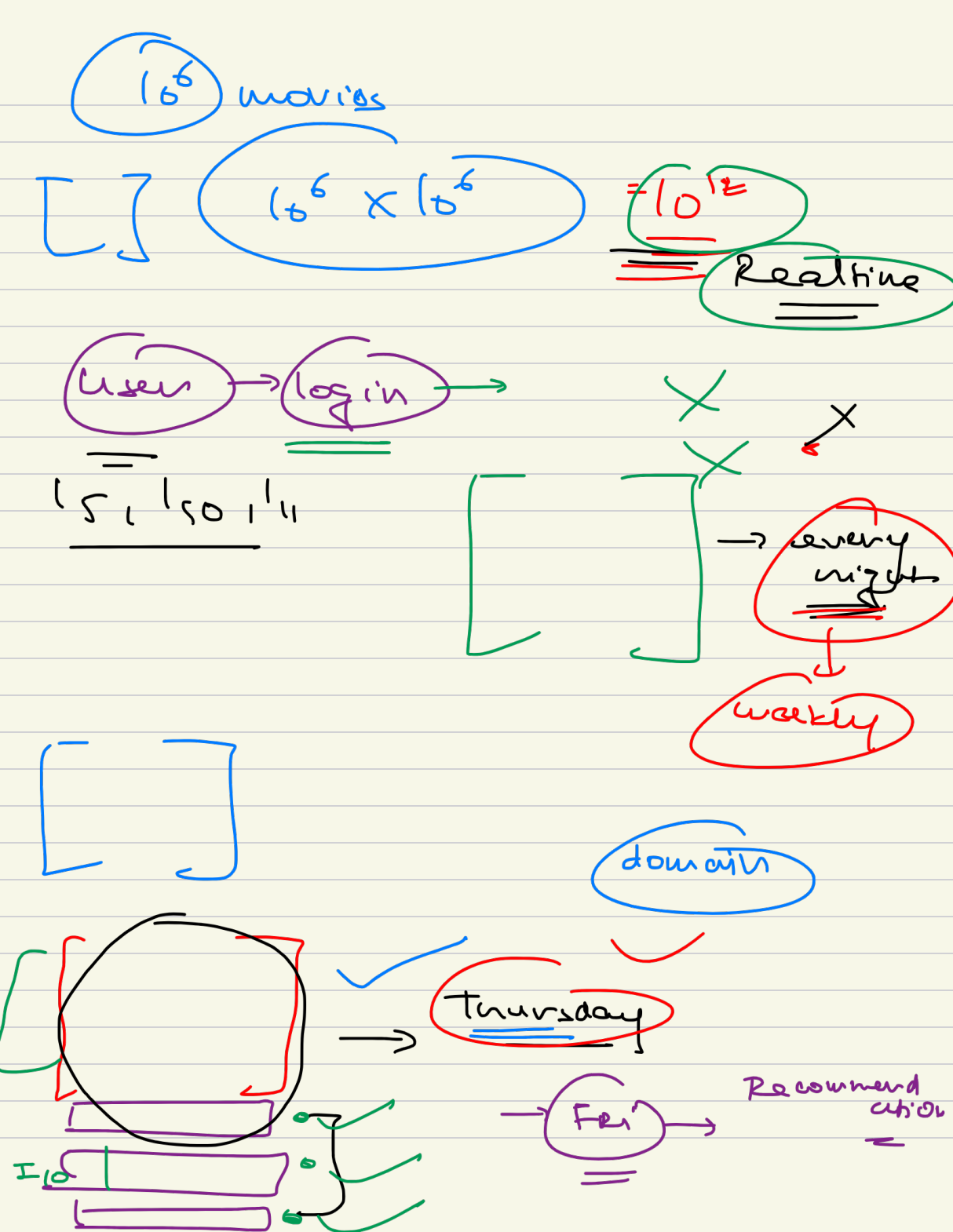
cos(I_2, I_3)

$I_3 \rightarrow$ [5 | 1]

I_2 [4 | 2]

cosine (I_3, I_2)

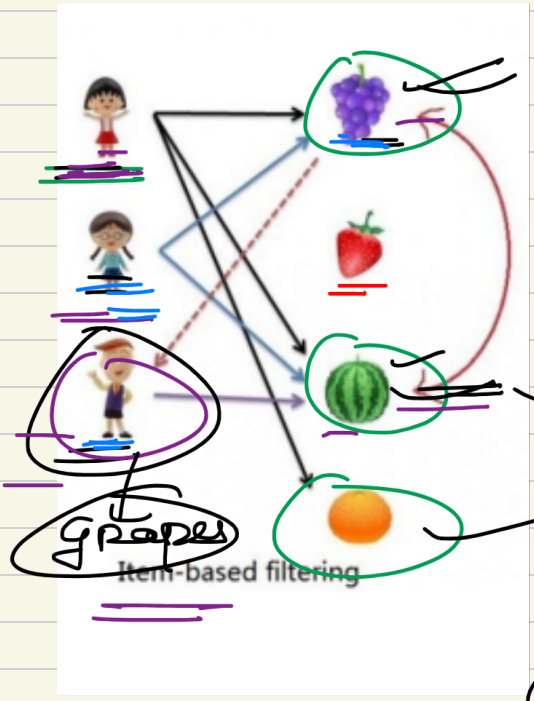
$= \frac{5 \times 4 + 1 \times 2}{\sqrt{26}\ \sqrt{20}} = ?$


10^6 movies
$10^6 \times 10^6$
$= 10^{12}$
Realtime
User
login
$10^5, 10^{50}, 11$
every night
weekly
domain
Thursday
Fri
Recommend
I_{10}

lookup ⇒ Highest

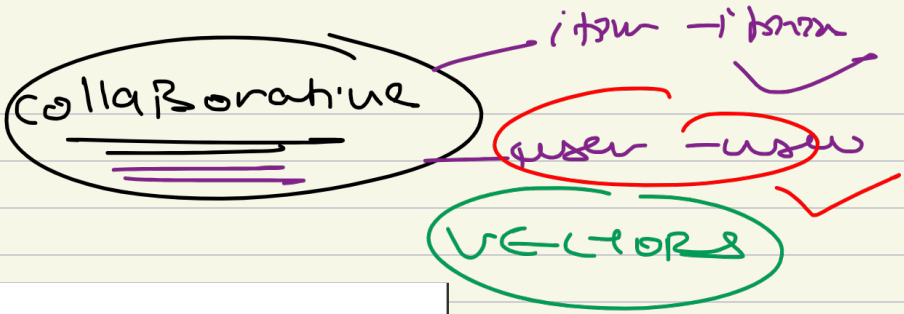
Recommend

(item item similarity)



[ ]

user – user ⇒ Next



Quiz time!

Time Left: 21s

Which statement is true about the calculation of the similarity matrix in platforms like Amazon?

10 users have participated

A	The calculation was performed whenever a user logged in	20%
B	The calculation was only performed when the user made a purchase	0%
C	The calculation was done during the nights or even weekly	60%
D	The calculation was instantaneous, thanks to high-speed servers	20%

End Quiz Now

